

YIFEI ZHANG

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EDUCATION

B.Sc. in Computer Science and Technology (FinTech)

Department of Computer Science and Technology, *Top Talent Class*

Nanjing University

Sep 2020 – Jun 2024

M.Sc. in FinTech and Financial Engineering

School of Management & Engineering, supervised by Prof. [Honghai Yu](#)

Nanjing University

Sep 2024 – Jun 2027

INTERNSHIP EXPERIENCE

Century Frontier (Quantitative Investment)

July 2026 – Present

Agentic Quantitative Research Intern

- Led the design and development of Century Frontier's **AutoResearch** system and **Knowledge Base**, introducing LLM agents into the quantitative investment research workflow.

XYZ AI Lab (Shanghai Xingyun Zhili AI Research Institute)

April 2026 – July 2026

Research Intern advised by [Xiao Yang](#) and [Jiang Bian](#)

- Core member of **Bounded AI4AI** for LLM post-training: led the design of the Agentic Search **evaluation harness** and the **SFT data recipe**, supporting XYZ-Aquila-mini's **SOTA among comparable model sizes** on seven Deep Search benchmarks, with pro achieving the highest publicly reported scores on three benchmarks; model weights, training data and runtime harness are all open-sourced (see Projects).

Microsoft Research Asia (MSRA) | Machine Learning Group

May 2025 – Jan. 2026

Research Intern advised by [Xiao Yang](#) and [Weiqing Liu](#)

- Core research intern on **RD-Agent (14,000+ Stars)**, with **two co-first-authored works**: led the design of the Gome algorithm and cross-model scaling experiments (**ACL Findings 2026**, **open-source SOTA** on MLE-Bench); led the construction of the FT-Dojo benchmark and the development of FT-Agent (**ICML 2026**, best on **10 of 13 tasks**, outperforming Claude Code and Codex); awarded MSRA "**Rising Star**" Outstanding Intern.

The Chinese University of Hong Kong, Shenzhen | School of Data Science

June 2024 – Mar. 2025

Research Intern advised by Prof. [Benyou Wang](#)

- Financial LLM evaluation and multi-agent social simulation: **led as co-first author** the UCFE Benchmark and TwinMarket, published at **NAACL Findings 2025** and **NeurIPS 2025**; received **Best Paper Award** at the ICLR Financial AI Workshop (1/53).

PROJECTS

A Controllable Self-Evolving R&D System for LLM Post-Training

April 2026 – July 2026

- Agentic Search Evaluation Harness**: co-built and open-sourced the Agent Runtime [AxisAgentic](#), and led the construction of a standardized evaluation harness covering seven Deep Search benchmarks. Under a controlled setting with fixed Model, Tool and Endpoint, the harness systematically ablates context-management and verification strategies such as retry, summary and self-verify, quantifying the **performance ceiling achievable by pure harness optimization** and improving the average score across the seven benchmarks by over **20 points**.
- Bounded AI4AI Paradigm and SFT Data Recipe Iteration**: co-designed and validated an R&D paradigm in which "**humans define the objective, exploration boundary and acceptance gates, while agents execute automated exploration and accumulate experience**". The pipeline automatically chains a four-stage closed loop of Researcher → Developer → Reviewer → Recorder, covering data filtering and synthesis, training and evaluation, and experience archiving, with support for parallel multi-experiment exploration and cross-validation reproduction. Unattended data-recipe iteration within the pre-defined gates improved the model's accuracy on an in-house evaluation set by **8%**.
- The above work supported the open release of **XYZ-Aquila-mini (35B)** and **Aquila-pro (397B)**: mini achieves **SOTA among comparable model sizes** on seven Deep Search benchmarks, and pro attains the highest publicly reported scores on BrowseComp-ZH, LiveBrowseComp and WideSearch. Model weights, training data and the runtime harness are all open-sourced.

Gome (RD-Agent, 14,000+ Stars) | First Author

May 2025 – Sept. 2025

- Observed that existing MLE agents, despite using different search topologies, process execution signals (error traces, training logs, code diffs) in the same way: they **compress them into a scalar score for candidate ranking, discarding the information about “which direction to change”**. Inspired by gradient-based optimization, Gome shifts the role of feedback from ranking to **update**: structured reasoning over execution signals guides the direction of code modification (analogous to gradients), while experience memory accelerates convergence (analogous to momentum) and multi-track parallelism broadens search coverage (analogous to distributed SGD).
- Achieves **35.1% any-medal rate (open-source SOTA)** under the MLE-Bench closed-world protocol. Across 10 models (GPT-4o-mini → GPT-5) compared against tree-search baselines, we identified a **crossover phenomenon**: tree search wins on weaker models while Gome dominates on stronger ones (**+7.1% on GPT-5**), revealing distinct scaling behaviors of the two paradigms as model capability grows.

FT-Dojo (RD-Agent, 14,000+ Stars) | Co-First Author

Sept. 2025 – Jan. 2026

- Built the first interactive benchmark for autonomous LLM fine-tuning, spanning **13 tasks across 5 domains** (mathematics, chemistry, finance, patents, and tabular QA), where an agent must independently complete the full loop of data filtering / cleaning / synthesis → training → evaluation → iterative diagnosis.
- Systematically uncovered three core bottlenecks of autonomous fine-tuning: context explosion, inefficient exploration due to the lack of fail-fast, and misreading of multi-level evaluation feedback. The proposed FT-Agent achieves **the best result on 10 of the 13 tasks, surpassing frontier baselines such as Claude Code and Codex**.

TwinMarket | Co-First Author

Oct. 2024 – Mar. 2025

- Led the design and development of a large-scale LLM-driven multi-agent simulation platform for financial markets. Each agent implements a perception–reasoning–action loop based on the BDI cognitive architecture, supporting scalable concurrent simulation with **1,000+ agents**.
- Designed an inter-agent social network communication mechanism, under which the system spontaneously exhibited collective intelligence phenomena such as **information cascades** and **herding effects** without explicit programming, reproducing stylized facts including fat-tailed return distributions and volatility clustering.
- Invited to give an oral presentation at the **ICLR 2025 Financial AI Workshop** (Singapore); received the **Best Paper Award** (1/53).

UCFE Benchmark | Project Lead

Aug. 2024 – Jan. 2025

- Designed and developed a user-centric evaluation framework for financial LLMs, constructing a dataset of 330 instances covering 17 task types through a survey of **804 participants**.
- Adopted a GPT-4o simulator with an **LLM-as-Judge** protocol, achieving strong alignment with human preferences (Pearson correlation **0.78**) and providing a reproducible automated evaluation pipeline for LLM capability assessment.

PUBLICATIONS (* INDICATES EQUAL CONTRIBUTION)

Reasoning as Gradient: Scaling MLE Agents Beyond Tree Search

Yifei Zhang*, Xu Yang*, Xiao Yang*, Bowen Xian, Qizheng Li, Shikai Fang, Jingyuan Li, Jian Wang, Mingrui Xu, Weiqing Liu, Jiang Bian.

ACL Findings 2026 (CCF-A)

FT-Dojo: Towards Autonomous LLM Fine-Tuning with Language Agents

Qizheng Li*, Yifei Zhang*, Xiao Yang*, Xu Yang*, Zhuo Wang, Weiqing Liu, Jiang Bian.

ICML 2026 (CCF-A)

TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets

Yuzhe Yang*, Yifei Zhang*, Minghao Wu*, Kaidi Zhang, Yunmiao Zhang, Honghai Yu, Yan Hu, Benyou Wang.

NeurIPS 2025 (CCF-A) & **Best Paper Award** at Financial AI @ ICLR 2025 (1/53).

UCFE: A User-Centric Financial Expertise Benchmark for Large Language Models

Yuzhe Yang*, Yifei Zhang*, Yan Hu*, Yilin Guo, Ruoli Gan, Yueru He, Mingcong Lei, Xiao Zhang, Haining Wang, Qianqian Xie, Jimin Huang, Honghai Yu, Benyou Wang.

NAACL Findings 2025 (CCF-B)

TECHNICAL REPORTS

R&D-Agent: An LLM-Agent Framework Towards Autonomous Data Science
Technical Report for the R&D-Agent Framework, 2025

AI4AI at Scale: A Full-Pipeline System for Enhancing LLM Agentic Capabilities
Technical Report for the New SOTA Deep Search Agent Developed by XYZ AI Lab, 2026

HONORS AND AWARDS (SELECTED)

Nanjing University National Scholarship (Top 0.2%)	<i>Oct. 2025</i>
Nanjing University Outstanding Graduate Student	<i>Nov. 2025</i>
Microsoft Research Asia Rising Star Outstanding Intern	<i>Jan. 2026</i>
China Moutai “Guozhi Dongliang” Excellence Talent Program (only 5 university-wide)	<i>May 2026</i>